1 mole of Fe^{2+} weighs 56g

2.686×10^{-3} moles of Fe^{2+} weighs $\left(\frac{56}{1} \times 2.686 \times 10^{-3}\right)$ grams

$= 0.1504g$

% purity of the iron wire $= \frac{0.1504}{0.1576} \times 100\%$

$= 95.4\%$

IODOMETRIC TITRATION

This titration involves the use of thiosulphate reactions with iodine solution. The indicator used here is starch solution. In this reaction the iodine is reduced to iodide ions and the thiosulphate is oxidized to tetrathionate ions.

The starch solution is used as an indicator to determine the end point of the titration. The starch solution is added at the end point when the solution turns pale yellow and the titration continued up to the end point.

Sodium thiosulphate exists as a clear crystals of $Na_2S_2O_3 \cdot 5H_2O$.

Sodium thiosulphate is a reducing agent but is not used as a primary standard because

(i) The water content of the crystals is variable.

(ii) The thiosulphate is easily decomposed by an acid to form a yellow precipitate of sulphur and sulphur dioxide gas. $S_2O_3^{2-}(aq) + 2H^+(aq) \rightarrow S(s) + SO_2(g) + H_2O(l)$

Therefore the solution of thiosulphate is never accurately

Half equations of the reaction between $S_2O_3^{2-}$ and Iodine

Oxidation reaction.

$2S_2O_3^{2-}(aq) \longrightarrow S_4O_6^{2-}(aq) + 2e$ ---(i)

Reduction reaction

$I_2(aq) + 2e \longrightarrow 2I^-(aq)$ ---(ii)

Overall equation for the reaction.

$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$

Note

The starch solution is added towards the end point of this titration to indicate the presence of iodine. The thiosulphate is run from the burette until the solution turns to pale yellow. At this point starch is added which changes the colour of the solution to blue. The titration is then continued until the solution turns to colourless and this marks the end point of the titration.

Example

Titration of thiosulphate with dichromate in the presence of potassium iodide

Here the dichromate oxidizes the iodide ions from potassium iodide to iodine in acidic conditions. The dichromate is reduced to chromium (III) ions and water.

The iodine produced is then titrated with sodium thiosulphate using starch solution as an indicator.

The end point is detected when the solution turns to blue-green.

Half equations

$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$ ---(I)

$2I^-(aq) \longrightarrow I_2(aq) + 2e$ ---(II)

Overall equation

Multiply equation (I) by 1 and equation (II) by 3 to obtain the overall redox reaction.

$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6I^-(aq) \longrightarrow 2Cr^{3+}(aq) + 7H_2O + 3I_2(aq)$

The iodine liberated reacts with thiosulphate

$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$

Qn1. A standard solution A was prepared by dissolving 1.185g of potassium dichromate(VI) and the resultant solution made up to 250 cm^3 with distilled water. 25 cm^3 of this solution on addition of sulphuric acid and potassium iodide solution required 17.5 cm^3 of sodium thiosulphate. Calculate the molarity of thiosulphate.

Soln

RFM of $K_2Cr_2O_7 = (39\times2) + (52\times2) + (16\times7) = 294$.

294g of $Cr_2O_7^{2-}$ are contained in 1mole.

1.185g of $Cr_2O_7^{2-}$ are contained in $\left(\frac{1}{294} \times 1.185\right)$ moles

$= 4.031\times10^{-3}$ moles of $Cr_2O_7^{2-}$.

250 cm^3 of solution contains 4.031×10^{-3} moles of $Cr_2O_7^{2-}$

25 cm^3 of solution contains $\left(\frac{4.031 \times 10^{-3}}{250} \times 25\right)$ moles of $Cr_2O_7^{2-}$

$= 4.031 \times 10^{-4}$ moles of $Cr_2O_7^{2-}$

$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6I^-(aq) \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l) + 3I_2(aq)$

1 mole of $Cr_2O_7^{2-}$ produced 3 moles of I_2

4.031×10^{-4} moles of $Cr_2O_7^{2-}$ produced $\left(\frac{3}{1} \times 4.031 \times 10^{-4}\right)$ moles of I_2

$= 1.21 \times 10^{-3}$ moles of I_2.

$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$

1 mole of I_2 reacted with 2 moles of $S_2O_3^{2-}$

1.21×10^{-3} moles of I_2 reacted with $\left(\frac{2}{1} \times 1.21 \times 10^{-3}\right)$ moles of $S_2O_3^{2-}$

$= 2.42 \times 10^{-3}$ moles of $S_2O_3^{2-}$

17.5 cm^3 of solution contains 2.42×10^{-3} moles of $S_2O_3^{2-}$

1000 cm^3 of solution contains $\left(\frac{2.42 \times 10^{-3}}{17.5} \times 1000\right)$ moles of $S_2O_3^{2-}$

$= \underline{0.1383M\ S_2O_3^{2-}}$

Reaction with iodate ions

In acidic condition, iodate ions undergoes reduction to form iodine which is then titrated with thiosulphate ions on addition of potassium iodide.

Half equation

Reduction reaction

$2IO_3^-(aq) + 12H^+(aq) + 10e \longrightarrow 6H_2O(l) + I_2(aq)$ ---(I)

Oxidation reaction

$2I^-(aq) \longrightarrow I_2(aq) + 2e$ ----(II)

Multiply equation (I) by 1 and equation (II) by 5 and then add the 2 equations

$2IO_3^-(aq) + 12H^+(aq) + 10I^-(aq) \longrightarrow 6H_2O(l) + 6I_2(aq)$

Finally it becomes

$IO_3^-(aq) + 6H^+(aq) + 5I^-(aq) \longrightarrow 3H_2O(l) + 3I_2(aq)$

Example

Q1. 3.2g of potassium iodate was dissolved to make a litre of solution. 25 cm^3 of this solution on addition dilute sulphuric acid followed by potassium iodide solution required 24.5 cm^3 of sodium thiosulphate using starch solution. Calculate the molar concentration of sodium thiosulphate.

soln

RFM of KIO_3 = (39x1) + (127x1) + (16x3) = 214.

214g of KIO_3 are contained in 1mole

3.2g of KIO_3 are contained in $\left(\frac{1}{214} \times 3.2\right)$ moles of KIO_3

= 0.015M KIO_3.

1000 cm^3 of solution contains 0.015 moles of IO_3^-

25 cm^3 of solution contains $\left(\frac{0.015}{1000} \times 25\right)$ moles of IO_3^-

= 3.75×10^{-4} moles of IO_3^-

$IO_3^-(aq) + 6H^+(aq) + 5I^-(aq) \longrightarrow 3H_2O(l) + 3I_2(aq)$

1mole of IO_3^- produced 3moles of I_2

3.75×10^{-4} moles of IO_3^- produced $\left(\frac{3}{1} \times 3.75 \times 10^{-4}\right)$ moles of I_2

= 1.125×10^{-3} moles of I_2.

$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$

1mole of I_2 reacted with 2 moles of $S_2O_3^{2-}$

1.125×10^{-3} moles of I_2 reacted with $\left(\frac{2}{1} \times 1.125 \times 10^{-3}\right)$ moles of $S_2O_3^{2-}$

= 2.25×10^{-3} moles of $S_2O_3^{2-}$.

24.5 cm^3 of solution contains 2.25×10^{-3} moles of $S_2O_3^{2-}$

1000 cm^3 of solution contains $\left(\frac{2.25 \times 10^{-3}}{24.5} \times 1000\right)$ moles of $S_2O_3^{2-}$

= 0.092M $S_2O_3^{2-}$.

Reaction with manganate (VII) ions

In acidic conditions manganate (VII) ions oxidize iodide ions from potassium iodide to iodine.

The iodine liberated can then be titrated against a standard solution of sodium thiosulphate using starch solution as an indicator.

Half equations

Reduction equation:

$$MnO_4^-(aq) + 8H^+(aq) + 5e \longrightarrow Mn^{2+}(aq) + 4H_2O \quad --(i)$$

Oxidation equation:

$$2I^-(aq) \longrightarrow I_2(aq) + 2e \quad ---(ii)$$

Overall equation:

Multiply equation (i) by 2 and equation (ii) by 5. Add the 2 half equations to obtain the overall redox reaction equation.

$$2MnO_4^-(aq) + 16H^+(aq) + 10I^-(aq) \longrightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 5I_2(aq)$$

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2$$

Qn. 1.55g of potassium dichromate was dissolved to make 250cm³ of solution. 25cm³ of this solution on addition of dilute sulphuric acid followed by potassium iodide required 17.5cm³ of sodium thiosulphate using starch indicator. Calculate the molar concentration of sodium thiosulphate

[K = 39, Cr = 52, O = 16]

Experiment I

You are provided with the following

DA_1 which is a solution containing $32.3gl^{-1}$ of sodium thiosulphate, $Na_2S_2O_3 \cdot 5H_2O$.

DA_2 which is potassium manganate (VII) solution of unknown concentration

0.5M potassium iodide solution

2M sulphuric acid solution.

Solid N, which is an impure ferrous ethanedioate, $(CO_2)_2Fe$.

Theory:

In acidic medium, manganate (VII) ions reacts with thiosulphate ions, ethanedioate ions and iron (II) ions according to the equation below:

$$2MnO_4^-(aq) + 10S_2O_3^{2-}(aq) + 16H^+(aq) \longrightarrow 2Mn^{2+}(aq) + 5S_4O_6^{2-}(aq) + 8H_2O(l)$$

$$2MnO_4^-(aq) + 5C_2O_4^{2-}(aq) + 16H^+(aq) \longrightarrow 2Mn^{2+}(aq) + 10CO_2(g) + 8H_2O(l)$$

$$MnO_4^-(aq) + 5Fe^{2+}(aq) + 8H^+(aq) \longrightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l).$$

Procedure I

(a) Pipette $25cm^3$ or $20cm^3$ of DA_2 into a clean conical flask, add $10cm^3$ of 2M sulphuric acid followed by $10cm^3$ of 0.5M potassium iodide solution and titrate with DA_1 from the burette until the end point when the solution becomes pale yellow. Add 4-5 drops of starch solution and continue with titration up to the end point. Repeat the titration until you obtain consistent results.

Enter your results in Table I below.

Table I

Volume of pipette used = $25.0cm^3$

Final burette reading (cm^3)	20.70	20.60	20.60
Initial burette reading (cm^3)	0.00	0.00	0.00
Volume of DA_1 used (cm^3)	20.70	20.60	20.60

Tire values used to calculate average volume of DA_1 = (20.60 and 20.60) cm^3

Average volume of $DA_1 = \frac{(20.60 + 20.60)cm^3}{2} = 20.60cm^3$

procedure II

(b) Weigh accurately about 1.3g of N and place it into a $250cm^3$ volumetric flask. To it add about $150cm^3$ of 2M sulphuric acid shake to dissolve and make up to a mark with distilled water. Label the resultant solution DA_3

(c) Pipetted 25cm³ or 20cm³ of BA_1 and heat to boiling temperature. Titrate the hot solution with BA_2 from the burette. Repeat the titration 2-3 times until you obtain consistent results. Enter your results in table II below.

Mass of empty container + N = 13.9g

Mass of empty container = 12.6g

Mass of N = 1.3g

Volume of pipette used = 25.0cm³

Table II

Final burette reading (cm^3)	20.50	19.50	20.50
Initial burette reading (cm^3)	0.00	0.00	0.00
Volume of BA_2 used (cm^3)	20.50	19.50	20.50

Titre values used to calculate average volume of BA_2 = (20.50 and 20.50) cm^3

Average volume of BA_2 = $\left(\frac{20.50 + 20.50}{2}\right) cm^3$ = 20.50cm³

(d) Calculate the

(i) Molar concentration of manganate (VII) ions in BA_2 [Na=23, S=32, O=16, H=1]

RFM of $Na_2S_2O_3 \cdot 5H_2O$ = (23x2) + (32x2) + (16x3) + (5x18) = 248.

248g of $S_2O_3^{2-}$ contains 1mole

22.3g of $S_2O_3^{2-}$ contains $\left(\frac{1}{248} \times 22.3\right)$ moles of $S_2O_3^{2-}$

= 0.0899M $S_2O_3^{2-}$

1000cm³ of solution contains 0.0899 moles of $S_2O_3^{2-}$

20.6cm³ of solution contains $\left(\frac{0.0899}{1000} \times 20.6\right)$ moles of $S_2O_3^{2-}$

= 1.852×10^{-3} moles of $S_2O_3^{2-}$

10moles of $S_2O_3^{2-}$ reacted with 2moles of MnO_4^-

1.852×10^{-3} moles of $S_2O_3^{2-}$ reacted with $\left(\frac{2}{10} \times 1.852 \times 10^{-3}\right)$ moles of MnO_4^-

= 3.704×10^{-4} moles of MnO_4^-

25cm³ of solution contains 3.704×10^{-4} moles of MnO_4^-

1000cm³ of solution contains $\left(\frac{3.704 \times 10^{-4}}{25} \times 1000\right)$ moles of MnO_4^-

= 0.0148M MnO_4^-

(ii) Moles of manganate (VII) ions that reacted with 25 cm^3 of A_3.

1000 cm^3 of solution contains 0.0148 moles of MnO_4^-

20.5 cm^3 of solution contains $\left(\frac{0.0148}{1000} \times 20.5\right)$ moles of MnO_4^-

$= 3.034 \times 10^{-4}$ moles of MnO_4^-.

(iii) Moles of iron (II) in 25 cm^3 of A_3.

1 mole of manganate (VII) ions reacted with 5 moles of Fe^{2+}

3.034×10^{-4} moles of MnO_4^- reacted with $\left(\frac{5}{1} \times 3.034 \times 10^{-4}\right)$ moles of Fe^{2+}

$= 1.517 \times 10^{-3}$ moles of Fe^{2+}.

(iv) Mass of $(CO_2)_2Fe$ in 250 cm^3 of A_3 [C = 12, Fe = 56, O = 16]

$(CO_2)_2Fe(s) \longrightarrow C_2O_4^{2-}(aq) + Fe^{2+}(aq)$

Mole ratio of $C_2O_4^{2-} : Fe^{2+} = 1:1$

Moles of $C_2O_4^{2-} = 1.517 \times 10^{-3}$ moles

25 cm^3 of solution contains 1.517×10^{-3} moles of $C_2O_4^{2-}$

250 cm^3 of solution contains $\left(\frac{1.517 \times 10^{-3}}{25} \times 250\right)$ moles of $C_2O_4^{2-}$

$= 0.01517$ moles of $C_2O_4^{2-}$.

RFM of $(CO_2)_2Fe = (12 \times 2) + (16 \times 4) + (56 \times 1) = 144$.

144 g of $C_2O_4^{2-}$ are contained in 1 mole

0.01517 moles of $C_2O_4^{2-}$ are contained in (144×0.01517) g

$= 2.184$ g.

(v) Percentage purity of $(CO_2)_2Fe$ in N.

$$\% \text{ purity} = \frac{\text{pure mass}}{\text{impure mass}} \times 100\%$$